

Jos Buttler T20I Runs Prediction Using Supervised Learning

A Machine Learning Approach for Predicting Cricket Player Performance

1. Selected Player

Jos Buttler (England): batter / wicketkeeper-batter. Chosen because he has a long, publicly verifiable T20I record with many measurable batting features.

2. A) Task / Objective

Predict how many runs Jos Buttler will score in his **next T20 International** match using historical T20I performance data.

Learning type	Supervised Learning	Problem type	Regression	Target	Runs
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3. B) Experience

The model learns from Jos Buttler's previous T20I match-by-match performances. Each past match is a **labelled example**: the match features are known and the actual runs scored are known, so the model can learn the link between them.

Runs (target)	Balls Faced	Strike Rate	Batting Position	Opposition
Venue	Fours	Sixes	Previous Match Runs	Recent Form

To avoid data leakage, a match's own runs, balls and strike rate are never used as inputs for that match; only earlier matches feed the history features.

4. C) Learning Domain

Domain: Cricket / Sports Analytics | **Task:** Player Performance Prediction | **Algorithm:** Decision Tree Regressor

Why Decision Tree? It is easy to understand and interpret, handles non-linear relationships, suits regression (each leaf predicts an average of runs), and is good for explaining the model during the viva. Random Forest may be used as an optional comparison.

Workflow of Learning

1	Identify the Problem	Predict Jos Buttler's runs in his next T20I match.
2	Data Collection	Collect his match-by-match T20I data (date, opposition, venue, batting position, runs, balls, strike rate, fours, sixes) from ESPNcricinfo, Cricsheet, Kaggle/GitHub.
3	Data Preprocessing	Clean the data: fix missing values, duplicates, wrong data types and invalid records.
4	Feature Engineering	Add new useful columns: Previous Match Runs, Last 3 and Last 5 Match Average. Remove unnecessary columns.
5	Model Selection	Choose Decision Tree Regressor.
6	Data Splitting	Older matches for training, newer matches for testing.
7	Model Training	Train the model to predict Runs from the features.
8	Evaluation & Prediction	Check accuracy using MAE, RMSE and R^2 (values to be calculated after training), then predict future runs.

Data Sources / References

- ESPNcricinfo: <https://www.espncriinfo.com/>
- ESPNcricinfo Statsguru: <https://stats.espncriinfo.com/>
- Cricsheet: <https://cricsheet.org/>
- Kaggle: <https://www.kaggle.com/>
- GitHub: <https://github.com/>

No Jos Buttler-specific dataset link was supplied; the exact dataset URL will be added as the primary reference once the dataset is chosen.